

TP73-supported TSS-local regulatory comparison

Scope

This report integrates the prior CD44, TGFB1, and SERPINE1 evidence on the same genomic axis. It uses every Ensembl Regulation feature overlapping a selected transcript-oriented **–500/+200 bp** TSS window. Adjacent selected windows are combined only for display; each feature is intersected with that displayed stretch before sequence comparison.

For each connected stretch, the report shows:

1. selected TSS windows and all overlapping Ensembl regulatory features;
2. the 15 candidate/cofactor JASPAR TFBS tracks as one shared layer;
3. the 12 TP73 CUT&RUN and H3K4me3 BigWig lanes from the prior report; and
4. one promoterome-recurrence strip per regulatory-feature intersection.

The similarity search compares those exact feature intersections with 389,722 distinct GRCh38/Ensembl-116 transcript-linked promoter windows. It requires at least 40 aligned bp, at least 80% identity, and $E \leq 1e-5$. Self-locus and same-gene targets are excluded. Counts are complete: the target cap was not reached.

Main observations

CD44

Four Ensembl feature intersections span the two selected TSS windows. The eMAR and two promoter intersections have no qualifying other-gene recurrence. Promoter ENSR11_9HXXD has two short matches—to MIR3201 and RSL24D1P9—but neither covers 25% of its 501-bp sequence.

TGFB1

The 740-bp clipped eMAR and 271-bp promoter have no qualifying other-gene recurrence. The 22-bp CTCF feature remains visible as biological context but is shorter than the declared fragment-similarity gate.

SERPINE1

The distal TSS stretch contains the only broad recurrence signal. A repeat-rich approximately 100-bp tract at **–334..–235 bp relative to TSS 101122158** lies within promoter ENSR7_93H5NS and its enclosing eMAR. The top ZNF586 promoter matches cover 28.9% of the 346-bp promoter intersection. Twelve other genes reach at least 25% coverage; none reaches 50%.

The adjacent enhancer ENSR7_BQ3T2 has only two short other-gene matches and no match covering 25%. In the proximal SERPINE1 stretch, the eMAR, two promoters, and enhancer have no qualifying recurrence. Its 22-bp CTCF feature is again motif-scale rather than a ≥ 40 -bp fragment query.

Reporter-design consequence

The distal SERPINE1 recurrent tract is the clearest boundary for an explicit deletion/split contrast. A reporter panel could preserve the local CUT&RUN and cofactor-TFBS context while comparing constructs that retain or omit this tract. The relatively sequence-restricted proximal features are useful contrasts, not automatically better promoters.

These results do **not** establish promoter activity, direct TP73 binding, or fragment sufficiency. Ensembl feature types are annotations, JASPAR sites are predictions, CUT&RUN is occupancy/enrichment evidence, and sequence recurrence is structural evidence. Reporter activity still requires experimental testing.